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PANEL DISCUSSION 5: INTEGRATING MACHINE LEARNING AND TRADITIONAL STATS IN YOUR RESEARCH: A DISCUSSION USING CROWDSOURCED VIGNETTES

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Machine learning (ML) is the process of developing systems that learn to detect, predict, and act based on data and experiences without being explicitly programmed. ML has many applications in behavioral science that include (but are not limited to) predicting behaviors and treatment outcomes, uncovering latent patterns within complex datasets, individualized treatment tailoring, and developing adaptive intervention tools. With access to increasing amounts of data collected via digital health tools (e.g., electronic health records, sensor systems, smartphones), many behavioral scientists are asking: "How do I know if ML is right for my research, and how does it compare to "traditional" statistical approaches (e.g., regressions)?" The answers to these questions are essential to the successful application of ML in behavioral research. However, the decision to employ ML, "traditional" statistics, or *both* must be carefully considered and is often dependent on multiple project-specific factors (e.g., research aims and design, study population, type and amount of data, team expertise).

This cross-disciplinary panel aims to provide insights into how experienced researchers make mindful decisions regarding the implementation of ML in their work. It is designed to facilitate engaging discussions about how to select analytic approaches that best fit different project aims. The panelists have previously used ML to classify and predict lapses from smoking cessation and weight management treatments, personalize in-the-moment interventions to increase physical activity, and detect stress and problematic eating from wearable sensors. The panel will compare the purposes of ML and "traditional" statistics, with an emphasis on how *both* can be employed in the same study. Panelists will then be given three 'vignettes' – crowdsourced from SBM members prior to the panel – that describe hypothetical research goals (e.g., to develop a just-in-time adaptive intervention that reduces lapses from a smoking cessation program). For each vignette, the panelists will discuss which types of research designs and accompanying data-analytic approaches they may consider using—both as part of the intervention and for post-study analyses—to meet the research goals. Throughout the discussion, they will highlight methodological or practical considerations that might impact their design decisions and how they may choose to implement ML solutions. In line with the conference theme of "moving behavioral science upstream", panelists will also be asked to comment on the upstream impacts of analytic decisions (e.g., developing generalizable models for real-world use, implementation of ML solutions) and how relevant social and structural determinants of health could be considered in research design and analysis.

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PANEL DISCUSSION 6: DESIGNING HEALTH: NOVEL APPROACHES TO THE USE OF VISUALS IN HEALTH MESSAGING

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Visuals can powerfully communicate health information, yet care must be taken when selecting visuals to meet the goals of communication efforts. This panel will explain how visuals can be used to convey complex health information, engage diverse audiences, facilitate meaningful decision-making processes, and promote positive behavior change. A focus of this panel will be more recent and novel approaches to using visuals to enhance health communication efforts and to improve health outcomes. The purpose of this panel is twofold: (1) to explore different types of visual messaging and (2) to discuss how to select visual content depending on message goals, as well as the decisional and behavioral contexts.

Our panel presents experts from various disciplines, including communication, public health, and medicine. Panelists will discuss the latest research on the topic and will share their insights on how visuals can be used in research and public health communication practice. Presenter 1 will provide an overview of current research in visuals and highlight how they provide a powerful opportunity to facilitate informed decision-making and reduce heuristics and biases. Presenter 2 will discuss their research on visual content effects. Presenter 3 will describe the use of visuals to more effectively communicate risk and will highlight the use of AI in co-designing visuals for risk messages. Presenters 4 will discuss a human-centered design study that includes the creation of journey maps to describe Black sexual minority men's experiences for HIV prevention to inform the co-delivery of alcohol interventions.

This session will be informative and engaging for anyone with interests in health communication, decision-making, and behavior change. Attendees will gain a deeper understanding of how novel approaches to creating visuals can be harnessed to communicate health information effectively and foster informed decision-making. We invite discussion, collaboration, and the exchange of innovative ideas to advance the study and practice of visual health communication.

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